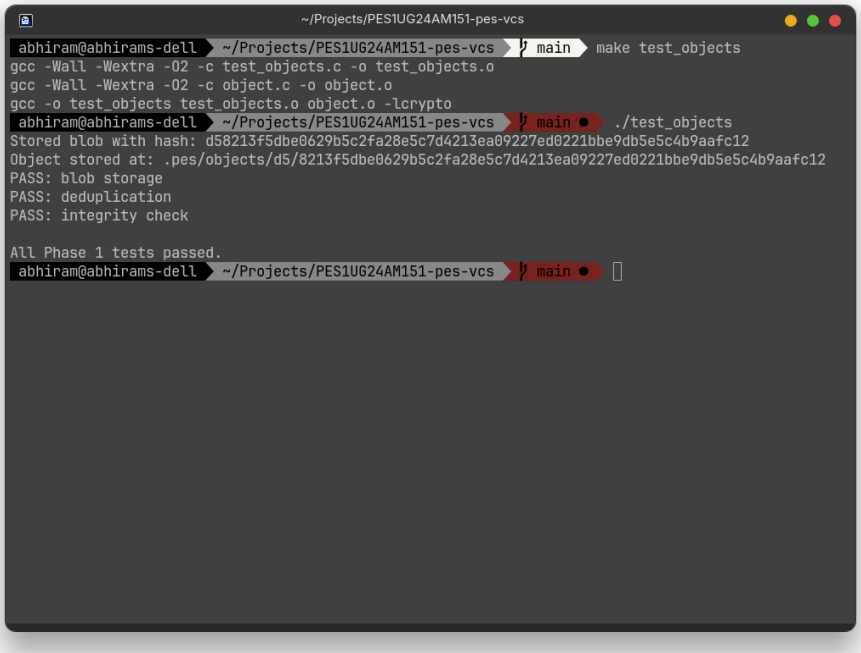
PES-VCS Lab Report

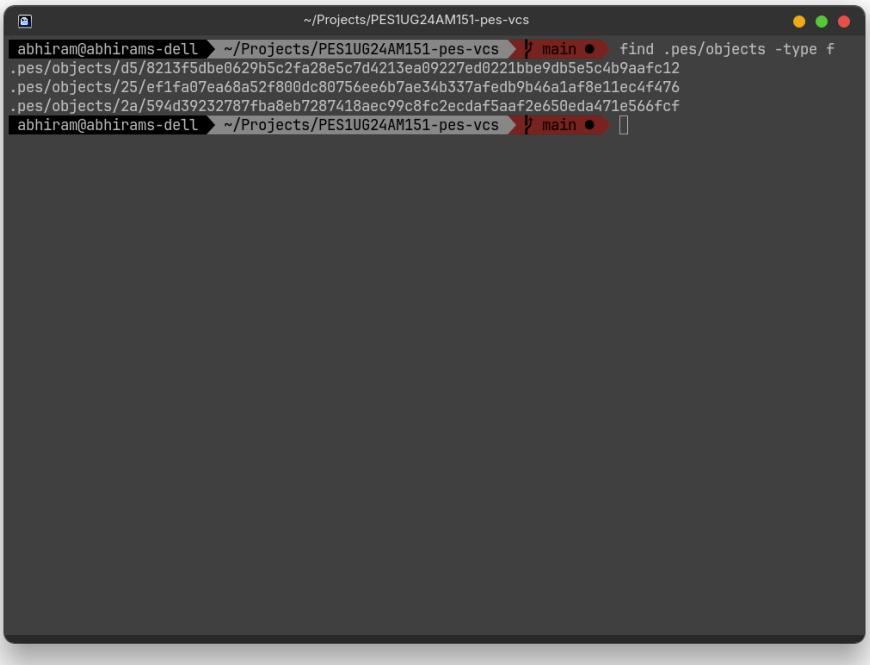
- **Name:** M RISHYENDRA
- **SRN:** PES1UG24AM154
- **Repository:** PES1UG24AM154-pes-vcs
- **Platform:** EndeavourOS (Arch Linux-based)
- **Github repo**: https://github.com/Rishyendra-12/PES1UG24AM154-pes-vcs

Phase 1 — Object Storage

Screenshot 1a

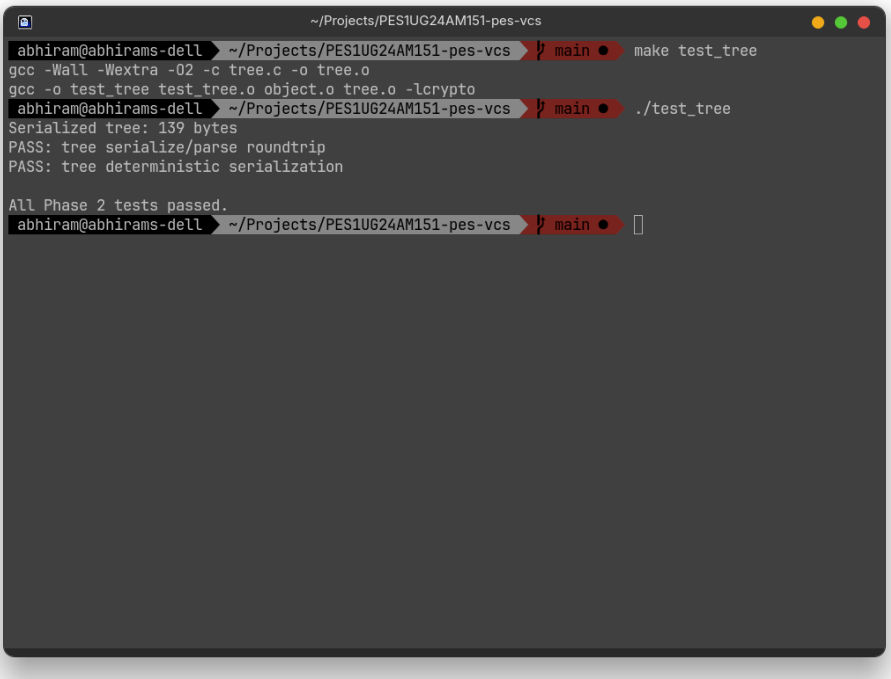


Screenshot 1b



Phase 2 — Tree Objects

Screenshot 2a



Screenshot 2b

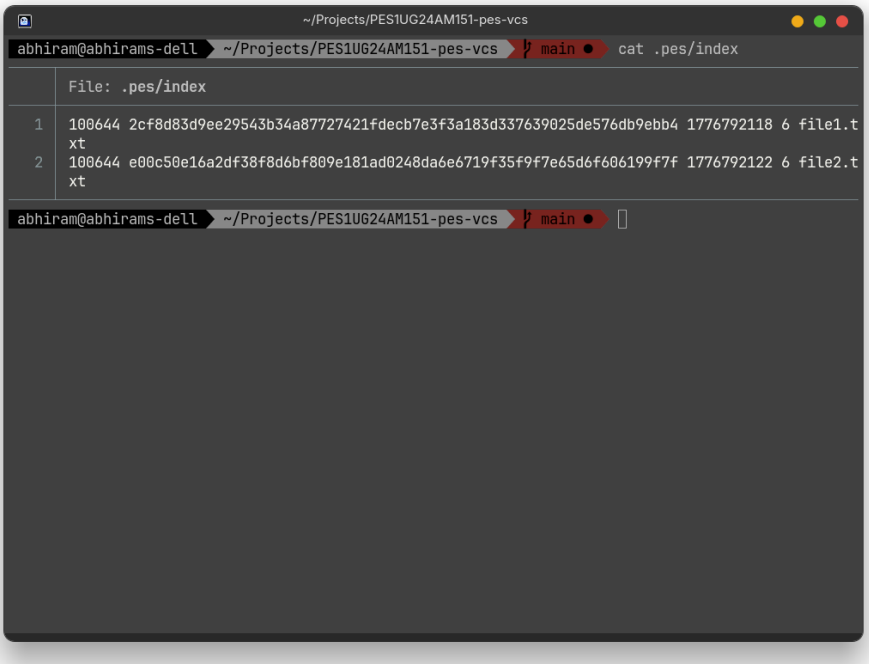
```
~/Projects/PES1UG24AM151-pes-vcs
abhiram@abhirams-dell  ~/Projects/PES1UG24AM151-pes-vcs  main ●  xxd .pes/objects/d5/8213f5db
e0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
abhiram@abhirams-dell  ~/Projects/PES1UG24AM151-pes-vcs  main ●
```

Phase 3 — Index / Staging Area

Screenshot 3a

```
~/Projects/PES1UG24AM151-pes-vcs
abhiram@abhirams-dell  ~/Projects/PES1UG24AM151-pes-vcs  main ●  echo "hello" > file1.txt
abhiram@abhirams-dell  ~/Projects/PES1UG24AM151-pes-vcs  main ●  echo "world" > file2.txt
abhiram@abhirams-dell  ~/Projects/PES1UG24AM151-pes-vcs  main ●  ./pes add file1.txt file2.tx
t
abhiram@abhirams-dell  ~/Projects/PES1UG24AM151-pes-vcs  main ●  ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  Makefile
  untracked:  README.md
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  test_objects
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h

abhiram@abhirams-dell  ~/Projects/PES1UG24AM151-pes-vcs  main ●
```

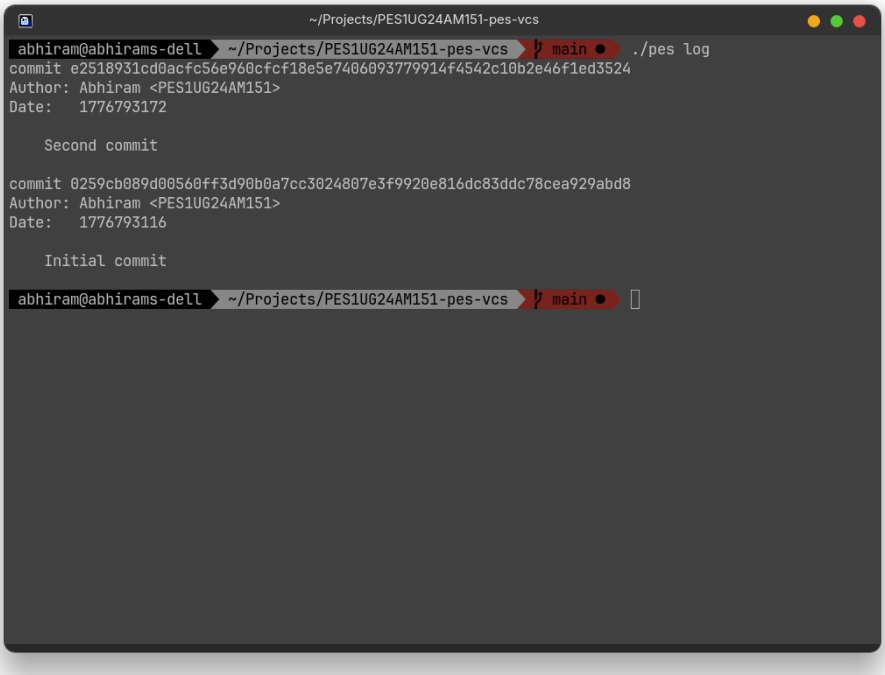
Screenshot 3b



Phase 4 — Commits and History

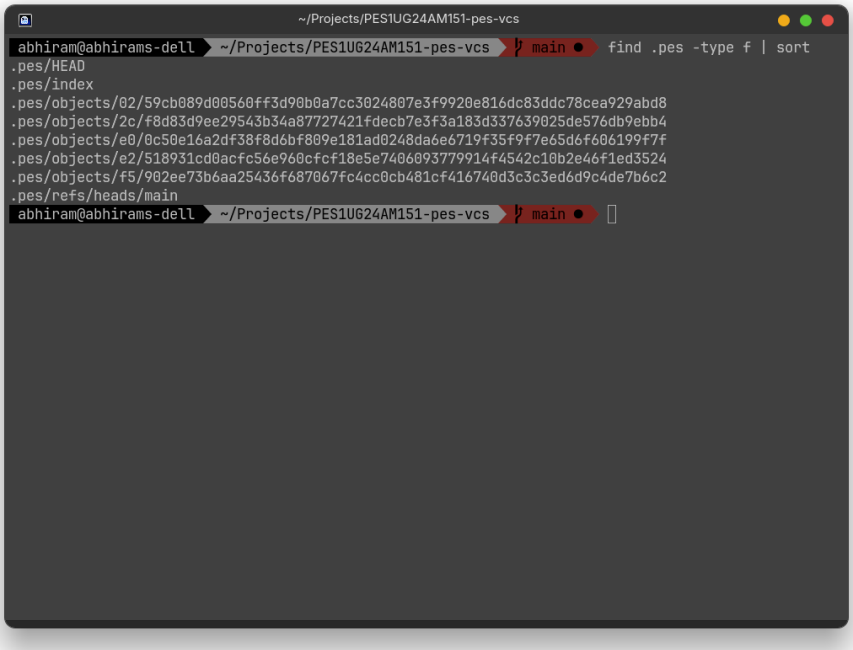
Screenshot 4a

`./pes log` showing three commits.



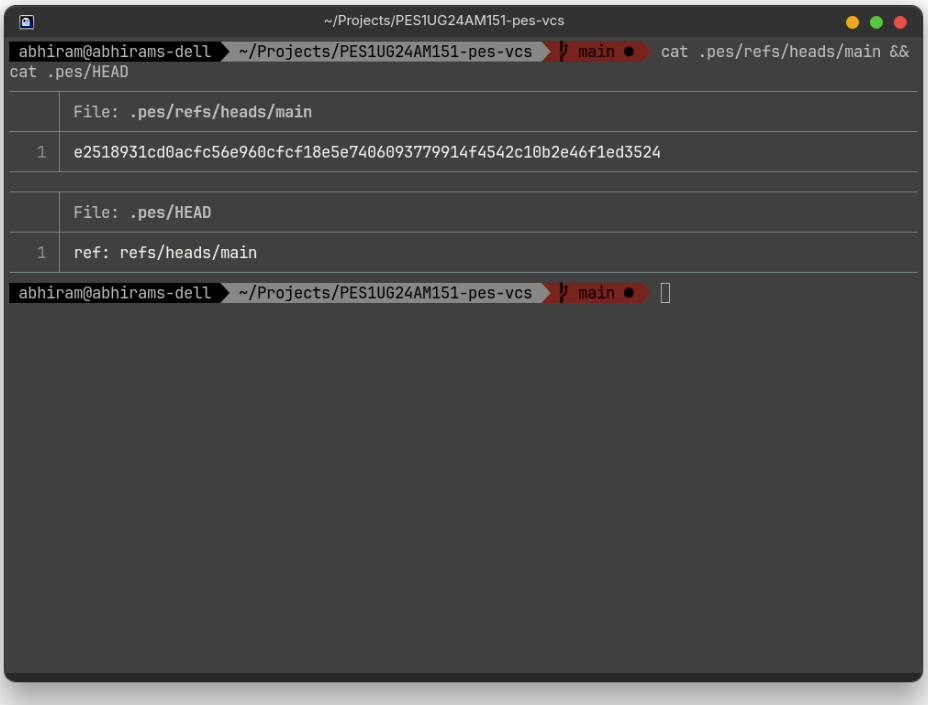
Screenshot 4b

`find .pes -type f | sort` showing object store growth after three commits.



Screenshot 4c

`cat .pes/refs/heads/main` and `cat .pes/HEAD` showing the reference chain.



Final Test

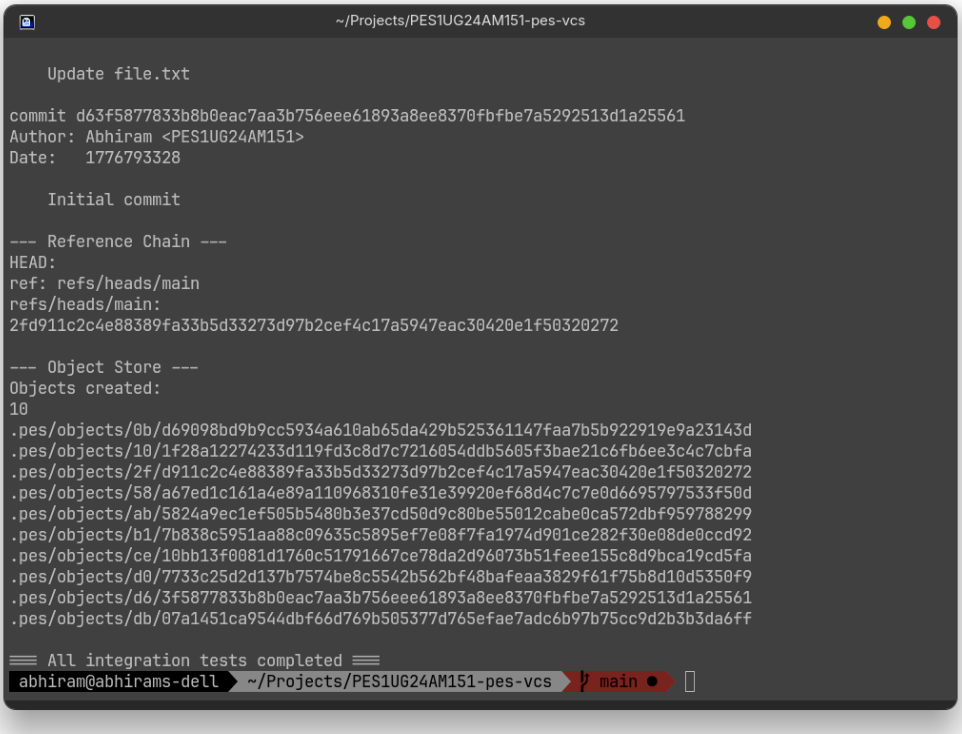
Final Test — Part 1

```
~/Projects/PES1UG24AM151-pes-vcs
abhiram@abhirams-dell  ~/Projects/PES1UG24AM151-pes-vcs  main ●  make test-integration
═══ Running integration tests ═══
bash test_sequence.sh
═══ PES-VCS Integration Test ═══

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: d63f5877833b... Initial commit

Log after first commit:
commit d63f5877833b8b0eac7aa3b756eee61893a8ee8370fbfbe7a5292513d1a25561
Author: Abhiram <PES1UG24AM151>
Date:   1776793328
```

Final Test — Part 2

```
~/Projects/PES1UG24AM151-pes-vcs
--- First Commit ---
Committed: d63f5877833b... Initial commit

Log after first commit:
commit d63f5877833b8b0eac7aa3b756eee61893a8ee8370fbfbe7a5292513d1a25561
Author: Abhiram <PES1UG24AM151>
Date:   1776793328

    Initial commit

--- Second Commit ---
Committed: ce10bb13f008... Update file.txt

--- Third Commit ---
Committed: 2fd911c2c4e8... Add farewell

--- Full History ---
commit 2fd911c2c4e88389fa33b5d33273d97b2cef4c17a5947eac30420e1f50320272
Author: Abhiram <PES1UG24AM151>
Date:   1776793328

    Add farewell

commit ce10bb13f0081d1760c51791667ce78da2d96073b51feee155c8d9bca19cd5fa
Author: Abhiram <PES1UG24AM151>
Date:   1776793328

    Update file.txt

commit d63f5877833b8b0eac7aa3b756eee61893a8ee8370fbfbe7a5292513d1a25561
Author: Abhiram <PES1UG24AM151>
```

Final Test — Part 3



Analysis Questions

Q5.1: Implementing `pes checkout <branch>`

To implement `pes checkout <branch>`, the repository must update `.pes/HEAD` and the branch reference under `.pes/refs/heads/<branch>`. If the checkout is to a normal branch, `HEAD` should continue pointing to the branch name rather than a raw commit hash. The working directory must then be rewritten so it matches the tree of the target commit: files that no longer exist on the target branch must be removed, files that differ must be replaced, and new files and directories from the target branch must be created.

This operation is complex because it must protect user data. Checkout cannot blindly overwrite tracked files if the user has uncommitted local changes. It also has to handle nested directories, deletions, file modes, and untracked files that might be clobbered by the switch.

Q5.2: Detecting dirty working directory conflicts

A dirty working directory conflict can be detected by comparing the current working directory, the index, and the target branch's tree. For each tracked path, compare the file in the working directory against the index metadata first. If size or modification time changed, the file may be dirty. For a stronger check, hash the working file and compare it against the blob hash stored in the index. Then compare the target branch's tree entry for that path with the current staged version.

If a tracked file is modified locally and the target branch has a different version of that same file, checkout should refuse. In other words, checkout must stop whenever switching branches would overwrite uncommitted changes.

Q5.3: Detached HEAD

Detached HEAD means `HEAD` stores a commit hash directly instead of a branch reference. Commits made in this state still create valid new commit objects, but no branch pointer is advanced to remember them. The new commits are reachable only through the direct `HEAD` history until the user switches away.

A user can recover detached-HEAD commits by creating a new branch at the commit hash, or by updating an existing branch reference to that hash. If the hash is still known, the commit history can be preserved without loss.

Q6.1: Garbage collection and space reclamation

Garbage collection should begin from all branch tips and from `HEAD` if it points directly to a commit. From those roots, the algorithm recursively traverses commit parents and commit trees, and then each tree's child blobs and subtrees. Every visited object hash is inserted into a hash set to avoid revisiting the same object multiple times.

After traversal, any object not in the reachable set can be deleted. A hash set is the right data structure because lookups are fast and duplicates are naturally eliminated. With 100,000 commits and 50 branches, the number of objects visited is approximately the number of unique reachable commits plus the trees and blobs they reference. Because branches usually share history, the traversal is closer to the size of the reachable graph than to 50 times the number of commits.

Q6.2: Why concurrent GC is dangerous

Running garbage collection concurrently with commit creation is dangerous because commit creation is multi-step. A commit may write blobs first, then trees, then the commit object, and

finally update the branch ref. If GC runs in the middle, it may see the new blob or tree as unreachable and delete it before the commit object or ref update makes it reachable.

Git avoids this with locking, atomic ref updates, and grace periods before pruning unreachable objects. Recently created or recently referenced objects are not deleted immediately, which prevents GC from racing against in-progress commits.

Submission Checklist

- Phase 1 screenshots added
- Phase 2 screenshots added
- Phase 3 screenshots added
- Phase 4 screenshots added
- Analysis questions answered
- Final integration test included